

Manuel Valle Torre

MSC COMPUTER SCIENCE · DATA SCIENCE AND TECHNOLOGY

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AI Engineer & Data Scientist with a strong background in Multi-Agent Systems, LLMs, and Behavioral Analytics. Proven expertise in the full engineering lifecycle—from designing ETL pipelines and architecting AI-driven web applications (JELAI) to deploying production-ready models. Experienced in analyzing complex user interaction data to drive real-time personalization and automated interventions.

Work Experience

Centre for Education and Learning - TU Delft

Delft, Netherlands

PHD RESEARCHER - AI & MULTI-AGENT SYSTEMS

Apr. 2021 - Today

- Architected and developed **JELAI**: an open-source **Multi-Agent System** environment integrating **LLMs** into Jupyter Notebooks to facilitate real-time user interaction tracking and AI-driven support (repo.)
- Engineered **Sequence Mining** and NLP pipelines to extract high-dimensional user behavioral patterns from interaction logs, utilizing these insights to build predictive models for real-time monitoring.
- Implemented **GenAI** solutions to analyze user problem-solving strategies, creating automated feedback loops that adapt to user intent in real-time.
- Led the technical design of Human-Centered AI tools, facilitating workshops with 30+ stakeholders to translate complex user requirements into iterative system improvements and engineering specifications.
- Supervised 6 Master's theses, guiding the implementation of impactful research prototypes involving **Transformers**, **RAG**, and dashboard engineering.

Best Practices Learning Analytics - Npuls

Netherlands

LEARNING ANALYTICS & AI CONSULTANT

Jan. 2024 - Today

- Advised stakeholders on **Generative AI** implementation strategies, translating complex Machine Learning research into actionable technical roadmaps for non-technical audiences and policymakers (link).
- Designed and delivered a technical workshop on “Best Practices for Generative AI and Dashboards” for international researchers at LAK25, focusing on integrating GenAI into existing data infrastructure. (link)
- Co-organized national knowledge exchange events, fostering a community of practice around data-driven decision-making and system architecture in the Dutch sector.

Centre for Education and Learning - TU Delft

Delft, Netherlands

RESEARCH ENGINEER

Jan. 2019 - April 2021

- Developed and deployed **ELAT**: a full-stack web application (Node.js/JavaScript) for processing massive log datasets. Engineered the architecture to parse, clean, and visualize complex event data, significantly reducing data preparation time. (demo)
- Designed and deployed scalable **ETL pipelines** to aggregate and compute daily user performance metrics from raw logs, supporting a study with 800+ users via a custom Canvas dashboard.
- Authored Python curriculum and implementation procedures for the “Digital Skills Project”, automating data export workflows and enhancing cross-faculty digital literacy.
- Enhanced the backend and frontend of **SearchX** (Information Retrieval system) by implementing user interaction tracking modules, vital for experiments in active reading and search behavior. (repo)

Web Information Systems - TU Delft

Delft, Netherlands

RESEARCH ENGINEER

Feb. 2018 - Jan. 2019

- Engineered **NLP pipelines** for **Named-Entity Recognition (NER)** and user-generated content analysis. Managed data collection, feature engineering, and model validation for detecting adverse patterns in text data.
- Co-developed technical course materials (lectures, labs) for CSE1500 Database Technology, teaching SQL and database management concepts to 500+ Computer Science students.

Skills

AI & ML

LLM Implementation (RAG, Agents), Generative AI, TENSORFLOW, NLP. Sequence Mining, Predictive Modeling, Deep Learning. PYTHON (Advanced).

Data & Eng

SQL, DOCKER, MONGODB, NEO4J. ETL Pipelines, Data Visualization, Experimental Design. GIT, Data Management.

Web & Dev

FASTAPI, FLASK, REACT, JAVASCRIPT, NGINX. R, RESTful APIs.

General

Academic & Technical Writing, User Behavior Analysis. Languages: Spanish (Native), English (C2), Dutch (Basic).

Selected Technical Publications

Systems & AI Research

Delft, Netherlands

FOCUS: NLP, MULTI-AGENT SYSTEMS, AND SYSTEM ARCHITECTURE

Recent Highlights

- *LLM Chatbots in High School Programming: Exploring Behaviors and Interventions*. Manuel Valle Torre, Marcus Specht, Catharine Oertel. (ACM-SAC 2026). (pdf) **[LLMs / User Behavior]**
- *JELAI: Integrating AI and Learning Analytics in Jupyter Notebooks*. Manuel Valle Torre, Thom van der Velden, Marcus Specht, Catharine Oertel. (AIED 2025). (pdf) (code) **[Multi-Agent Systems]**
- *The Sequence Matters in Learning - A Systematic Literature Review*. Manuel Valle Torre, Catharine Oertel, Marcus Specht. (LAK 2024). (pdf) **[Sequence Mining]**
- *ELAT: An open-source, privacy-aware and browser-based log data analysis tool*. Manuel Valle Torre, Esther Tan, Claudia Hauff. (LAK 2020). (pdf) (demo) **[Full Stack / Analytics]**
- *Training Data Augmentation for Detecting Adverse Drug Reactions in User-Generated Content*. Sepideh Mesbah, Jie Yang, Robert-Jan Sips, Manuel Valle Torre, Christoph Lofi, Alessandro Bozzon, Geert-Jan Houben. (EMNLP 2019). (pdf) **[NLP / Data Augmentation]**
- *TSE-NER: An Iterative Approach for Long-Tail Entity Extraction in Scientific Publications*. Sepideh Mesbah, Christoph Lofi, Manuel Valle Torre, Alessandro Bozzon, Geert-Jan Houben. (ISWC 2018). (pdf) **[NER / Knowledge Graphs]**

Technical Mentorship

Delft University of Technology

Delft, Netherlands

MSC THESIS SUPERVISOR (FOCUS: APPLIED AI & ENGINEERING)

Apr. 2021 - Today

- Using LLMs for Support in High School Programming Education. T. van der Velden (pdf) **[LLMs / Prompt Engineering]**
- Enhancing Programming Education Through LLM-Based Learning Analytics. C.B.Z. Busropan (pdf) **[GenAI / RAG]**
- Automated Writing Feedback using Transformers. P.J. van der Laan (pdf) **[Transformers / BERT]**
- Computational Thinking Dashboards: For learners in Jupyter notebooks. B. Agarwal (pdf) **[Data Visualization]**

Education

Delft University of Technology

Delft, Netherlands

MSC COMPUTER SCIENCE - DATA SCIENCE & TECHNOLOGY

Sep. 2015 - Jan. 2018

- Thesis: *Shared Perspectives for Query by Example in E-Commerce Systems*. Engineered a personalized search engine using **Word2Vec embeddings** and **Clustering** to map user intent from textual reviews. (code)
- Specialization: Focused on Data Science, Web Data Engineering, and User Behavior Analysis, with core subjects in Data Management, (Cyber) Data Analytics, Experiment Design, and Web Information Systems.

Monterrey Institute of Technology and Higher Education

San Luis Potosí, México

BSC BUSINESS INFORMATICS

Aug. 2010 - Feb. 2015

- Focus on Information Systems Engineering and Business Process Management. (GPA: 9.2/10)